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Unraveling the nexus among achievement goals, self-efficacy, and motivation regulation and their joint effects on L2 student writers' writing achievement: a person-based approach

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Abstract
It is unknown how achievement goals, self-efficacy, and motivation regulation are interrelated and how they together influence L2 writing achievement. The current study adopted a person-based approach and explored the nexus among L2 student writers' achievement goals, self-efficacy, and motivation regulation and the joint effects on their writing achievements. A total of 416 tertiary students from Southwest China were recruited as voluntary participants. Their achievement goals, writing self-efficacy, and motivation regulation strategies were measured with the Writing Achievement Goals Scale, the Genre-Based L2 Writing Self-Efficacy Scale, and subscales of the Writing Strategies for Self-Regulated Learning Questionnaire, respectively. Their writing achievements was measured with two tasks: argumentative and narrative writing. Latent profile analyses revealed four profiles of achievement goals: "Low All Goals", "Mastery High", "Average All Goals", and "High All Goals". ANOVA and Welch's Test results further showed that the participants assigned to these four profiles differed in writing achievement goals, writing self-efficacy, motivation regulation strategies, and writing achievement in the narrative task. Additionally, multiple regression analyses revealed that their profiles varied in the predictive effect of writing self-efficacy and motivation regulation strategies on writing achievements. Methodological and pedagogical implications were also discussed.

Keywords Profiles of achievement goals · Motivation regulation strategies · Writing self-efficacy · Person-based approach

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Introduction

Writing is widely conceived as a challenging and complex task, characterized by linguistic, cognitive, and metacognitive challenges (i.e., Graham et al., 2005; Hashemian & Heidari, 2013; Troia et al., 2012). These challenges become even more pronounced in the context of second/foreign language (L2) writing, where learners must grapple not only with the inherent difficulties of writing but also with the additional burden of limited language proficiency and unfamiliar rhetorical conventions (Yang & Zhang, 2023). To overcome these compounded challenges, student writers must maintain and bolster their motivation, ensuring sustained effort and interest in writing tasks (Manchón et al., 2007; Teng & Zhang, 2020; Zhang, 2022; Zhang & Zhang, 2019; Zhang & Zhang, 2023; Zimmerman & Risemberg, 1997). Consequently, understanding how to strengthen and sustain motivation among L2 student writers is essential for improving their writing outcomes.

Motivation research, a broad field (Wigfield & Eccles, 2002), has identified key context-specific constructs, such as achievement goals and self-efficacy beliefs (Huang, 2016; Jiang et al., 2014; Zhang, 2022). Writing self-efficacy, defined as a writer's belief in their ability to successfully complete writing tasks (Pajares, 2003), has been consistently linked to writing performance in both first language (L1) and L2 contexts (e.g., Zhang & Zhang, 2023; Zhang et al., 2025). Similarly, writing achievement goals, which relate to the reasons behind engaging in writing activities (e.g., Zhang, 2023), have gained considerable attention, with mastery goals and performance-approach goals showing positive effects on writing achievements (e.g., Limpo & Alves, 2017; MacArthur et al., 2016; Zhang, 2023) and writing behaviors (e.g., Lee & Evans, 2019) and commitment (e.g., Zhang, 2023). Although person-based research using cluster and latent profile analyses has yielded insightful classifications of achievement goal profiles (e.g., Conley, 2012; Paul et al., 2021; Schwinger & Wild, 2012; Tuominen-Soini et al., 2008, 2011), such investigations have been predominantly limited to L1 contexts (e.g., Camacho et al., 2023; Paul et al., 2021). This gap is notable, as the unique challenges of L2 writing necessitate a tailored exploration of motivational profiles to better inform instructional practices for this population.

Research in L1 writing has also documented a significant relationship between achievement goals and self-efficacy (e.g., Chea, 2012; Huang, 2016; Wolters, 2004), suggesting that learners who feel more competent tend to adopt mastery and performance-approach goals, while those with lower perceived competence are more inclined toward avoidance-oriented goals (Elliot, 1999). In L1 contexts, studies have shown that mastery and performance-approach goals positively influence writing self-efficacy, whereas performance-avoidance goals are negatively associated (Yilmaz Soylu et al., 2017). However, these interrelations have rarely been examined in L2 writing, leaving a critical gap in our understanding of how these constructs function when language proficiency challenges intersect with motivational dynamics.

Moreover, motivation regulation, a critical component of self-regulated learning (SRL), involves consciously managing motivation to achieve academic goals (Boekaerts & Cascallar, 2006; Pintrich, 2004; Wolters, 2003). Given the additional challenges confronted by L2 student writers, effective motivation regulation becomes paramount to maintain engagement and overcome writing obstacles (Teng & Zhang, 2018). From a social-cognitive perspective, SRL involves the interplay between personal factors (e.g., cognitive, affective, and biological events), behavior, and environmental influences (Bandura, 1999). Within this

framework, achievement goals and self-efficacy are fundamental to the adoption of motivation regulation strategies (Zimmerman, 1989). Unraveling the nexus among achievement goals, self-efficacy, and motivation regulation strategies is critical to understanding how to motivate these students to strengthen and sustain their perseverance and persistence in the L2 writing process. However, to our best knowledge, no studies have examined the aforementioned relationship and effects within L2 writing contexts.

To narrow down the lacuna in the available literature, the current study was designed to examine the relationship among achievement goals, self-efficacy, and motivation regulation strategies in L2 student writers by employing the person-based approach and also the predictive effects on L2 writing achievement. In doing so, the study addresses important gaps in understanding how to connect motivational beliefs (i.e. achievement goals and self-efficacy) and motivation regulation among L2 student writers and offers novel theoretical and pedagogical insights that can guide teachers in developing targeted interventions to support L2 student writers.

Literature review

Achievement goals and their profiles

Achievement goal orientations, one of the popular lines of inquiry into achievement goals, are concerned with the reasons behind individual learners' engagement in certain academic activities. Dweck and Leggett (1988) initially put forward two distinctive constructs of goal orientation from the social-cognitive perspective: learning or mastery goal orientation and performance goal orientation. The latter was further split into performance-approach and performance-avoidance goal orientations (Elliot & Church, 1997), thus formulating the trichotomous model of achievement goals. Mastery-oriented learners may focus on attaining a deep understanding of knowledge, learning new knowledge, and acquiring new skills and competence to improve themselves. In contrast, performance-approach-oriented learners may tend to maximize favorable judgment of their knowledge and competence, while performance-avoidance-oriented student writers tend to avoid unfavorable evaluations of certain competence or failures in their performance.

The trichotomous model has been widely adopted across different domains (e.g., Elliot & Murayama, 2008; Kaplan et al., 2009; Limpo & Alves, 2017) and has also been applied within L2 writing contexts. In L2 writing, mastery-oriented student writers may focus on attaining a deep understanding of writing, learning new writing knowledge, and acquiring new writing skills and competence to improve themselves. In contrast, for performance-approach-oriented student writers, favorable judgments of their writing competence are the focus, while performance-avoidance-oriented student writers tend to avoid unfavorable evaluations of certain writing competence or failures in writing achievement. Recent research (e.g., Zhang, 2023) revealed that mastery goals and performance-approach goals can enhance L2 student writers' perseverance of effort but performance-avoidance goals showed reverse effects and that perseverance of efforts acted as a mediator in the relationship between writing achievement goals and L2 writing achievement.

Studies have demonstrated mastery and performance-approach goals could facilitate individual learners in enhancing their academic achievements in different fields (e.g., Huang,

2011; Hulleman et al., 2010; Linnenbrink-Garcia et al., 2008; Maehr & Zusho, 2009; Middleton & Midgley, 1997; Senko et al., 2011). Accordingly, advocates of the multiple-goal perspective argued that endorsing mastery goals together with performance-approach goals simultaneously could bring greater benefits and gains to learners (e.g., Barron & Harackiewicz, 2001; Harackiewicz et al., 1998, 2002; Senko et al., 2011). Scholars have utilized a person-based approach to detect various potential combinations or profiles of mastery goals, performance-approach goals, and performance-avoidance goals along with work-avoidance goals (i.e., concerned with the absence of an achievement goal) among students learning various subjects (e.g., Conley, 2012; Niemivirta et al., 2019; Paul et al., 2021; Schwinger & Wild, 2012; Tuominen-Soini et al., 2008, 2011). In a synthetic review of the available studies on profiles of achievement goals, Wormington and Linnenbrink-Garcia (2017) summarized 10 profiles of achievement goals: Mastery High, Approach High, High All Goals, Performance-Approach High, Work Avoidance High, Average All Goals, Performance Low, Low All Goals, Performance-Appproach Low, and Performance/Work-Avoidance Low.

Paul et al. (2021) identified three profiles of writing achievement goals among the L1 student writers: Low On All, Average On All, and High On All. Additionally, they found that the three profiles were significantly different as regards their writing anxiety and their perceived use of revision strategies. Given that L2 student writers are often confronted with more significant challenges, such as varying levels of language proficiency, limited vocabulary, less familiarity with L2 writing convention, all of which may affect their achievement goal orientations in writing tasks, it is vital to investigate whether L2 student writers exhibit unique achievement goal profiles that could have distinct implications for instructional design and learner support.

Self-efficacy and achievement goals in L1 and L2 writing

Self-efficacy, defined as the belief in one's capabilities to execute tasks successfully (Pajares, 2003), is widely acknowledged as a crucial factor influencing performance. Writing self-efficacy is a multidimensional construct, including self-efficacy for convention, self-efficay for self-regulation, and self-efficacy for ideation (e.g., Bruning et al., 2013). In both L1 and L2 writing contexts, higher levels of writing self-efficacy have been linked to improved writing performance (Zhang & Zhang, 2023; Zhang, et al., 2025). Researchers have also suggested that achievement goals shape learners' self-assessments: students who feel highly effecicacious tend to adopt mastery and performance-approach orientations, whereas those with lower self-efficacy may lean toward avoidance orientations (Elliot, 1999; Huang, 2016).

Empirical studies in L1 writing have confirmed these associations. For instance, Pajares et al. (2000) found that mastery goals were positively correlated with self-efficacy and self-efficacy for self-regulation, performance-approach goals also had positive correlations with self-efficacy, and performance-avoidance goals demonstrated negative correlations with self-efficacy and self-efficacy for self-regulation. Later, Pajares and Cheong (2003) further found that learners possessing higher self-efficacy and self-efficacy for self-regulation tend to form higher mastery goals than those with lower self-efficacy. Recently, Limpo and Alves (2017) revealed that mastery goals impinged positively on different dimensions of writing self-efficacy (i.e., self-efficacy for convention, ideation, and self-regulation) and also mediated the effect of beliefs in skill malleability on self-efficacy for self-regulation. Additionally, Yilmaz Soylu et al. (2017) identifies the mediating roles of self-efficacy for conven-

tion in the relationship between performance goals and writing achievments. Despite these findings, the interplay between achievement goals and self-efficacy in L2 writing remains underexplored. Given the added challenges in L2 writing, it is critical to examine whether similar patterns hold and how these constructs interact in environments where language proficiency might alter their dynamics.

Links among achievement goals, self-efficacy, and motivation regulation strategies

Social cognitive theory postulates that individuals are autonomous beings capable of exerting an impact on their surroundings while concurrently being subject to the effects of external stimuli (Bandura, 1999, 2006, 2018). Within this framework, SRL involves the use of strategies to regulate their cognition, motivation, and behavior to attain their academic objectives, relying on their own beliefs in their ability to succeed (e.g., Pintrich, 2000; Zimmerman, 2000). Within SRL, motivation regulation, the deliberate processes by which learners sustain or enhance their motivation (Pintrich, 2000; Wolters, 2003), play a pivotal role in managing academic challenges. This is particularly true in L2 writing, where student writers are confronted with a large amount of challenges, as argued above.

Approaching from the social-cognitive perspective, Zimmerman and Risemberg (1997) postulated that a high level of self-regulation is crucial for student writers to become proficient and suggested a close relationship between writing self-regulation and writing self-efficacy. In the learn-to-write process, student writers are required to enhance and uphold their motivation by deploying a series of strategies to sustain their persistence and interest (e.g., Manchón et al., 2007; Teng & Zhang, 2022; Wang et al., 2025; Zhang & Zhang, 2019; Zimmerman & Risemberg, 1997). In the context of L2 writing, Teng and Zhang (2018) argued that motivation regulation strategies are prominently essential for L2 student writers due to the additional challenges and setbacks they might be confronted with in mastering L2 writing skills. Motivation regulation strategies refers to specific strategies that learners employ to maintain their motivation during the writing process, including interest enhancement, motivational self-talk, and emotional control (Teng & Zhang, 2018).

Motivation and self-regulation strategies should be integrated in the specific ways to facilitate students' engagement in academic task, as postulated by Kaplan et al. (2009). Furthermore, it is also argued that different achievement goals would necessitate student writers' employment of various strategies. Wolters and Rosenthal (2000) found that learning/mastery goal orientation and performance goal orientation exerted the predictive impact on the use of motivation regulation strategies. In L2 writing, Teng (2021) found that intrinsic goal orientation and self-efficacy significantly predicted motivation regulation strategies among L2 student writers.

As discussed before, achievement goals and self-efficacy are essential constructs in the domain of motivation, and they are closely correlated. Exploring the nexus between achievement goals, self-efficacy, and motivation regulation strategies in L2 writing may offer a nuanced understanding of how L2 student writers' motivation influence their L2 writing performance, which may facilitate teachers to tailor their instruction to motivate L2 student writers in the learn-to-write process, thus fostering autonomous learners who can regulate their writing motivation independently and effectively and achieving long-term success in L2 writing and overall L2 acquisition. However, no studies have yet examined

the interaction between achievement goals, self-efficacy, and motivation regulation strategies in L2 writing.

The present study

To bridge these gaps in the literature, the current study was designed to explore the connections among achievement goals, self-efficacy, and motivation regulation strategies and their effects on L2 student writers' writing performance through the person-based rather than the variable-based approach. The preference for the person-based approach over the variable-based one is ascribed to the convenience and efficiency with which the former could facilitate us to explore the similarities and differences among L2 student writers formulating profiles of different achievement goals and examine the nuances of the relationships among achievement goals, self-efficacy, and motivational self-regulation in L2 writing (e.g., Niemivirta et al., 2019; Wormington & Linnenbrink-Garcia, 2017) and their effects on L2 student writers' writing achievement. Specifically, the current study intended to answer the following research questions:

1) What are the profiles of L2 student writers' writing achievement goals?
2) Are there differences in writing self-efficacy and motivation regulation strategies across the profiles of achievement goasl in L2 student writers?
3) Are there differences in writing achievements in argumentative and narrative tasks across the profiles of achievement goals?
4) Do the predictive effects of writing achievement goals, writing self-efficacy, and motivation regulation strategies on writing achievements vary across those profiles?

Methods

Participants

The present study recruited a convenience sample of 436 English as a foreign language students from two medium-ranking universities in Southwest China. These students voluntarily participated in the study after assigning formal consent. The participants were selected after obtaining authorization from academic department heads, who disseminated recruitment announcements to students who had expressed interest through departmental administrative staffs. The students started learning English as a foreign language at least since junior middle school, with Mandrin Chinese as their first language. Among the participants, 57.34% ($n=250$) were female, and 42.66% ($n=186$) were male. The age of the participants ranged from 18 to 22, with an average age of 20.4. They were enrolled in various programs, including electronic engineering ($n=71$, 16.28%), computer science ($n=74$, 16.97%), education ($n=75$, 17.2%), business ($n=114$, 26.14%), and administration ($n=103$, 23.62%). The participants were in different academic years, with 47.25% ($n=206$) being freshmen, 25.46% ($n=111$) being sophomores, and 27.29% ($n=119$) being juniors.

Instruments

The participants' achievement goals, self-efficacy, and motivation regulation strategies were respectively gauged by three self-report scales: the Writing Achievement Goals Scale, the Genre-Based L2 Writing Self-Efficacy Scale, and the Scale for Motivation regulation Strategies. Additionally, their writing achievements was measured by two writing tasks, one argumentative and the other narrative.

The writing achievement goals scale

The Writing Achievement Goals Scale, developed by Yilmaz Soylu et al. (2017), was adapted here to assess the participants' writing achievement goals: mastery goals (4 items, for instance, "When I'm writing in my English class, I'm trying to become a better writer."), performance-approach goals (4 items, for instance, "When I'm writing in my English class, I'm trying to impress my teacher with my writing."), and performance-avoidance goals (4 items, for instance, "When I'm writing in my English class, I'm trying to hide that I have a hard time writing."). This self-report tool utilized a 5-point Likert scale, ranging from 0 (does not describe me) to 4 (describes me perfectly). It was validated in a study involving L1-English secondary learners, demonstrating strong psychometric properties. Since the scale was initially developed and validated in the L1 context, the wording of its items was carefully reviewed and modified to suit the L2 context. Furthermore, all scale items were translated into Chinese to ensure participants' full comprehension and minimize potential misunderstandings. The accuracy and equivalence of the translation were verified through a process of translation and back-translation. Additionally, the scale underwent further statistical tests to evaluate its psychometric properties among the targeted L2 student writers. The reliability of this scale was assessed using Cronbach's alpha coefficient, which yielded a value of 0.840, indicating acceptable internal consistency.

The genre-based L2 writing self-efficacy scale

The Genre-Based L2 Writing Self-Efficacy Scale was directly adapted from Zhang et al. (2023) and was specifically designed to assess the participants' writing self-efficacy: linguistic self-efficacy (5 items, for instance, "I can write a simple English sentence with grammatical structure."), self-regulatory self-efficacy (3 items, for instance, "I can realise my goal to improve my English writing."), classroom performance self-efficacy (4 items, for istance, "I can get an excellent grade for writing in the English course."), and genre-based performance self-efficacy (4 items, for instance, "I can write an English narrative that includes several things that happened to the characters."). Linguistic self-efficacy refers to students' belief in their own ability to access vocabulary, formulate sentences from their thoughts, and fulfill the demands of discourse; self-regulatory self-efficacy refers to students' confidence in their capacity to consciously direct and monitor their learning throughout the writing process; classroom performance self-efficacy centers on learners' belief in their capability to finish a specific writing assignment within an academic setting; genre-based performance self-efficacy is concerned with learners' confidence in their ability to complete the writing assignments with different genres. This instrument employed a self-report approach, utilizing a 5-point Likert scale ranging from 1 (not indicative of me) to 5 (perfectly descriptive of

me). The scale's reliability and validity were established through its validation with college L2 learners, demonstrating robustness in these aspects. To ensure accurate comprehension and minimize potential misunderstandings, all the items were translated into Chinese for the participants, with the translation's accuracy and equivalence verified through a process of translating and back-translating. Furthermore, the scale's reliability was assessed using Cronbach's alpha coefficient, which yielded a value of 0.920, indicating a high level of internal consistency.

The scale for motivation regulation strategies

Participants' employment of motivation regulation strategies was measured using the subscales of the Writing Strategies for Self-Regulated Learning Questionnaire designed by Teng and Zhang (2016), which covered three facets of motivation regulation strategies: Interest enhancement (4 items, for instance, " I look for ways to bring more fun to the learning of writing."), motivational self-talk (8 items, for instance, "I remind myself about how important it is to get good grades in writing courses."), and emotional control (3 items, for instance, "I tell myself not to worry when taking a writing test or answering questions in writing courses."). It was a domain-specific self-report instrument, with a 7-point Likert scale ranging from 1 (not at all true of me) to 7 (very much true of me). It was validated with L1-Chinese L2 learners from various majors, such as Economics, Civil Engineering, Law, and Psychology, demonstrating sound psychometric properties. The reliability of this scale was also assessed by Cronbach's alpha coefficient, whose value was 0.928, suggesting the sound internal consistency of this scale.

The writing test

Based on the provided prompts, participants were instructed to produce two compositions with a minimum word count of 150. Each writing task was to be completed within a 40-minute timeframe using an online writing platform, Pigai. While argumentative writing is widely utilized to evaluate writing achievements, we argue that relying solely on this genre fails to present a comprehensive evaluation of students' writing abilities, especially in a foreign or second language. Thus, incorporating writing tasks with two genres has the potential to facilitate a more complete assessment of the participants' writing achievements. The topics for both tasks were selected from the established item pool of the College English Test, Band 4 (CET 4), which demonstrated strong validity and reliability. The writing prompt for the argumentative task would ask the participants to take a stance on the impact of technology on education in China's university settings. The aim of this prompt is to argue either in favor of or against the assertion that computers and the Internet have enhanced both the speed and the standard of learning for students at the tertiary level. In contrast, the writing prompt for the narrative task would invite the participants to reflect on and share a significant moment or series of moments in their journeys of learning English. This could be an experience that was particularly challenging, rewarding, or transformative. The purpose of this prompt is to allow the participants to express how this experience impacted their relationship with English, what they learned from it, and how it has influenced their ongoing English learning. Further details can be found in Appendix A.

The ESL Composition Profile by Jacobs et al. (1981) was utilized to evaluate the participants' writing achievements. This analytic scoring rubric is widely recognized in L2 writing studies and has gained global popularity due to its user-friendly nature. It is commonly employed to assess the writing proficiency/achievement of L2 student writers. The profile assesses five dimensions of L2 writing achievement: content, organization, language, vocabulary, and mechanics. Each dimension is assigned a specific weight: content (30%), organization (20%), language (25%), vocabulary (20%), and mechanics (5%). These dimensions are rated on a scale of four levels, which include excellent to very good, good to average, fair to poor, and very poor.

Procedure

The study consisted of two phases, as shown in Fig. 1. During phase one, participants were instructed to complete the Writing Achievement Goals Scale, The Genre-Based L2 Writing Self-Efficacy Scale, and subscales of the Writing Strategies for Self-Regulated Learning Questionnaire. Prior to responding to the scales, participants were required to sign a consent form. The scales were completed using an online questionnaire platform called Wenjuanxing (also known as Questionnaire Star). Participants were informed that their responses

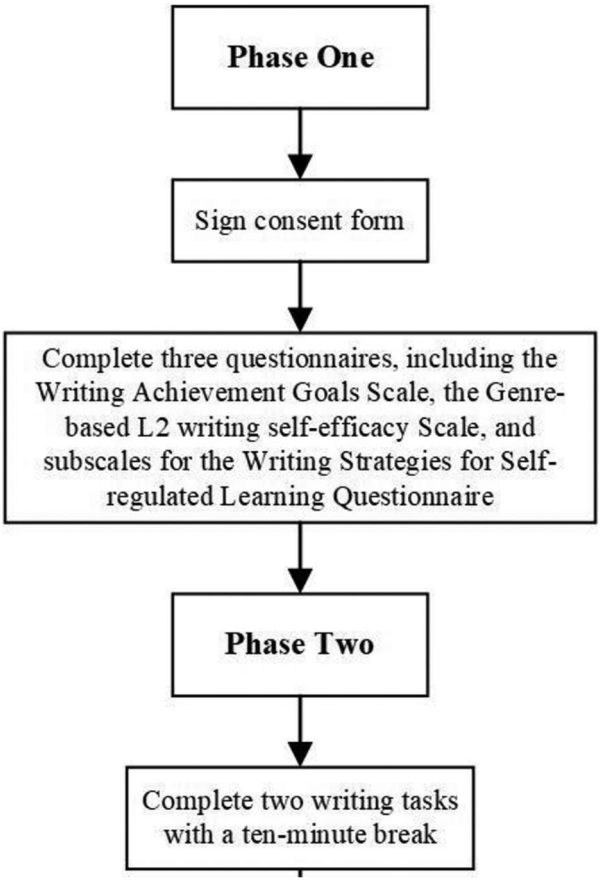


Fig. 1 Procedure for data collection

would not be evaluated as right or wrong based on specific scoring criteria. It would be greatly appreciated if they could kindly provide honest answers, as their feedback is highly valued. It is important to note that their responses would not affect their course grades in any way. Any questions or remarks they have would be acknowledged and addressed during and after completing the survey. On average, it took participants 4 to 7 min to finish the questions.

During the second phase, participants were instructed to complete an argumentative and a narrative writing assignment and were asked to complete each within a time limit of 40 min, with a ten-minute break between two tasks. During each session, they kept writing, no matter whether they reached the specified number of 150 words. A total of 832 English writing essays were collected for the purpose of evaluating the participants' writing achievements. The average text length of the collected essays in the argumentative task was 161 words, while that in the narrative task was 173 words. These essays were thoroughly and independently assessed by two qualified CET4 writing raters, who utilized the grading criteria established by Jacobs et al. Before scoring the collected essays, the raters spent one day familiarising themselves with the scoring rubric and reached a consensus on understanding the description of the grades in each dimension. The inter-rater reliability for the argumentative task was $r_{AB} = 0.939$, while for the narrative task, it was $r_{AB} = 0.897$, signifying consistent and dependable scoring. The final score on each writing task was calculated by averaging the scores given by each individual rater. The final scores on the argumentative task ranged from 49 to 89.25, while those on the narrative task varied from 48 to 90.5.

Data analysis

The data set underwent initial screening and cleaning procedures, following the recommendations of Dörnyei and Taguchi (2010), before any analysis took place. During this process, twenty responses displaying contradictory answers were removed from the database. Additionally, missing data were examined and cross-checked using the Statistics Package for Social Science (SPSS) 25. After completing the screening and cleaning steps, a total of 20 participants were excluded from the study, leaving 416 participants for the final analysis. We conducted exploratory latent profile analyses to discern L2 student writers' writing achievement goal profiles with the aid of Mplus 8.3 (Muthen & Muthen, 2017). As a common practice, the maximum likelihood estimation with robust standard errors was utilized to estimate the parameters of these newly constructed latent profiles. Furthermore, the estimates recommended by Jung and Wickrama (2008) were employed to evaluate the best-fitted profile of writing achievement goals: the Akaike information criterion (AIC), the Bayes Information criterion (BIC), the adjusted Bayes Information criterion (aBIC), and entropy. Besides, the differences between these profiles were evaluated by utilizing the Lo–Mendell–Rubin likelihood ratio test (LMR-LRT) and bootstrapped likelihood ratio test (BLRT).

Moreover, L2 student writers' differences in writing achievement goals, writing self-efficacy, and motivation regulation strategies across the best-fitted profiles of writing achievement goals were examined by employing Analysis of Variance (ANOVA) or Welch's Test by the mean of SPSS 25, and the differences between each profile pair were tested using post-hoc Scheffe's Tests where the *p*-value are adjusted. The choice of methods for examining the differences is dependent on the results of Levene's Test for Equality of Variance. Furthermore, the same statistical analyses were employed to detect L2 student writers' differ-

ences in writing achievements across the best-fitted profiles of writing achievement goals. Besides, the differences among the profiles in predictive effects of writing achievement goals, writing self-efficacy, and motivation regulation strategies on writing achievements were scrutinized by conducting multiple regression analyses on different profiles.

Results

Profiles of L2 student writers' writing achievement goals

Table 1 presents the descriptive data of the involved variables in the current study, including mean, standard deviation, Kurtosis, and Skewness. Additionally, it also presents the correlation matrix regarding the variables involved, which were tested by Pearson correlational analysis.

The exploratory latent profile analysis was employed to identify the profiles of writing achievement goals that L2 student writers formulated. Specifically, four solutions were tested: two-profile, three-profile, four-profile, and five-profile. The best-fitted latent profile might be determined by taking into account all the aforementioned estimates: higher entropy, lower AIC, BIC, and aBIC, and the significant p values for LMR and BLRT. In addition, the identified profiles were examined to ensure that they were conceptually distinct on the indicator variables. The model fit indices of four profile solutions are presented in Table 2.

The statistics in Table 2 indicate that the p-values for LMR for the two-profile and four-profile solutions were significant ($p_{\text{profile–2}} = 0.0002$, $p_{\text{profile–4}} = 0.0003$), whereas those for the three-profile and five-profile ones were non-significant ($p_{\text{profile–3}} = 0.3996$, $p_{\text{profile–5}} = 0.318$). These results may suggest that there were statistically significant differences in the one-profile versus two-profile and three-profile versus four-profile comparisons rather than in the two-profile versus three-profile and four-profile versus five-profile comparisons. Meanwhile, the entropy value for the four-profile solution is higher than those for the two-profile, three-profile, and five-profile solutions ($\text{entropy}_{\text{profile–4}}=0.929 > \text{entropy}_{\text{profile–3}} = 0.920 > \text{entropy}_{\text{profile–5}} = 0.813 > \text{entropy}_{\text{profile–2}} = 0.557$), indicating that the four-profile solution was the best-fitted one for L2 student writers' writing achievement goals profile. Additionally, with reference to achievement goals profiles identified in Wormington and Linnenbrink-Garcia (2017), the four-profile solution was conceptually reasonable. The nuance of the four-profile is shown in Fig. 2.

Figure 2 reveals that the learners assigned in Profile 1 were characterized by lower scores in mastery goals, performance-approach goals, and performance-avoidance goals; the learners in Profile 2 by higher mastery goals, medium performance-approach goals, and lower performance-avoidance goals; the learners in Profile 3 by medium scores in mastery goals, performance-approach goals, and performance-avoidance goals, the learners in Profile 4 by higher scores in mastery goals, performance-approach goals, and performance-avoidance goals. To keep consistent with the labels in previous studies, we named Profile 1 as "Low All Goals (LAGs) ", Profile 2 as "Mastery High (MH)", Profile 3 as "Average All Goals (AAGs)", and Profile 4 as "High All Goals (HAGs)".

The differences in three facets of writing achievement goals across four identified profiles were examined using ANOVA (or Welch's Test) and post-hoc Scheffe's Test, before

Table 1 Descriptive data and correlational matrix of the involved variables

	Mean	SD	Kurtosis	Skewness	1	2	3	4	5	6	7	8	9	10	11
1. MG	3.831	0.577	−0.564	2.323											
2. PApG	3.076	0.780	−0.123	−0.053	0.434**										
3.PAvG	3.066	0.814	−0.246	−0.374	0.126*	0.456**									
4. LS	4.247	1.099	−0.384	−0.097	0.317**	0.325**	0.064								
5. SRS	4.447	0.976	−0.102	0.306	0.452**	0.395**	0.120*	0.560**							
6. CPS	3.488	1.175	0.16	−0.359	0.241**	0.462**	0.238**	0.492**	0.490**						
7. GPS	4.132	1.052	−0.341	0.039	0.327**	0.377**	0.156**	0.533**	0.553**	0.642**					
8. IE	4.843	1.087	−0.497	0.172	0.521**	0.376**	0.152**	0.297**	0.489**	0.283**	0.281**				
9. MST	5.144	0.930	−0.666	0.943	0.644**	0.388**	0.140**	0.357**	0.549**	0.267**	0.330**	0.653**			
10. EC	4.846	1.073	−0.493	0.266	0.585**	0.374**	0.094	0.349**	0.518**	0.294**	0.297**	0.557**	0.703**		
11. SAT	75.191	6.925	−0.54	0.912	0.029	0.056	0.012	0.180**	0.107*	0.09	0.138**	0.012	0.017	0.039	
12. SNT	73.043	5.889	−0.51	1.328	0.155**	0.072	0.098*	0.09	0.086	0.031	0.066	0.121*	0.144**	0.107*	0.304**

Note. MG=mastery goals, PApG=performance-approach goals, PAvG=performance-avoidance goals, LS=linguistic self-efficacy, SRS=self-regulatory self-efficacy, CPS=classroom performance self-efficacy, GPS=genre-based performance self-efficacy, IE=interest enhancement, MST=motivational self-talk, EC=emotional control, SAT=score of argumentative task, SNT=scores of narrative task

**p<.01, *p<.05*

Table 2 Model fit indices for latent profile analyses

C	K	AIC	BIC	aBIC	Entropy	LMR	BLRT
2	10	2614.999	2655.306	2623.573	0.557	0.0002	0.0000
3	14	2563.123	2619.553	2575.127	0.920	0.3996	0.0000
4	18	2515.15	2587.703	2530.584	0.929	0.0003	0.0000
5	22	2454.571	2543.246	2473.434	0.813	0.318	0.0000

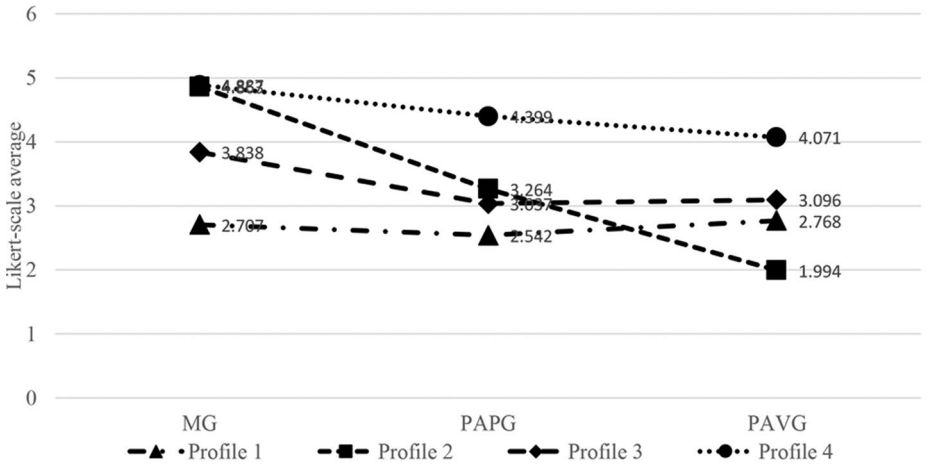


Fig. 2 Four profiles of L2 student writers' writing achievement goals. Note. MG = master goals, PAPG = performance-approach goals, PAVG = performance-avoidance goals

which Levene's Test for Equality of Variances was performed, as shown in Tables 3 and 4. The results of ANOVA or Welch's Test in Table 3 indicate that four profiles were significantly different in master goals ($F = 554.575$, $p < .001$, $\eta^2 = 0.766$), performance-approach goals ($F = 35.925$, $p < .001$, $\eta^2 = 0.207$), and performance-avoidance goals ($F = 34.23$, $p < .001$, $\eta^2 = 0.2$). The results of Scheffe's Test in Table 4 show that all profile pairs were significantly different in mastery goals, performance-approach goals, and performance-avoidance goals except the following pairs: the MH profile versus the HAGs profile pair in mastery goals ($p = .939$), the LAGs profile versus the AAGs profile pair in performance-approach goals ($p = .09$), and the MH profile versus the AAGs profile pair in performance-approach goals ($p = .681$).

Besides, the distribution of the participants in our sample was presented as follows: forty-one learners were assigned to the LAGs profile, accounting for 9.86%; twenty learners to the MH profile, accounting for 4.81%; 331 learners to the AAGs profile, accounting for 79.57%; and twenty-four learners to the HAGs profile, accounting for 5.77%.

Profile differences in writing self-efficacy and motivation regulation strategies

L2 student writers' differences in writing self-efficacy and motivation regulation strategies across the identified latent profiles of writing achievement goals were also examined utiliz-

Table 3 Results of differences in writing achievement goals, writing self-efficacy, and motivation regulation strategies across latent profiles

Factors	Mean				Levene(df)	F(df)	η^2
	LAGs	MH	AAGs	HAGs			
Writing achievement goal							
MG	2.656	4.863	3.836	4.917	10.546(3, 412)	Welch's F(3,50.563) = 554.575	0.766
PApG	2.616	3.225	3.028	4.406	0.757(3,412)	ANOVA F(3,412) = 35.925	0.207
PAvG	2.787	1.925	3.096	4.094	1.111(3, 412)	ANOVA F(3,412) = 34.23	0.2
Writing self-efficacy							
LS	3.639	4.460	4.240	5.192	0.386(3, 412)	ANOVA F(3,412) = 11.087	0.075
SRS	3.732	5.117	4.422	5.458	1.201(3, 412)	ANOVA F(3,412) = 22.043	0.138
CPS	3.110	3.450	3.459	4.573	1.647(3, 412)	ANOVA F(3,412) = 8.769	0.06
GPS	3.500	4.513	4.120	5.052	3.089(3, 412)	Welch's F(3,45.501) = 10.578	0.086
Motivation regulation strategies							
IE	3.805	5.638	4.831	6.125	1.798(3, 412)	ANOVA F(3,412) = 33.566	0.196
MST	4.021	6.188	5.136	6.302	4.423(3, 412)	Welch's F(3,50.563) = 64.313	0.294
EC	3.642	5.867	4.856	5.917	1.344(3, 412)	ANOVA F(3,412) = 39.968	0.225

Note. MG = mastery goals, PApG = performance-approach goals, PAvG = performance-avoidance goals, LS = linguistic self-efficacy, SRS = self-regulatory self-efficacy, CPS = classroom performance self-efficacy, GPS = genre-based performance self-efficacy, IE = interest enhancement, MST = motivational self-talk, EC = emotional control, LAGs = Low All Goals, MH = Mastery High, AAGs = Average All Goals, HAGs = High All Goals

ing ANOVA or Welch's Test. Similarly, Levene's Test for Equality of Variances was performed before ANOVA or Welch's Test. The results are shown in Table 3.

The results of ANOVA and Welch's Test in Table 3 indicate that four profiles were significantly different in four dimensions of writing self-efficacy: linguistic self-efficacy ($F = 11.087$, $p < .001$, $\eta^2 = 0.075$), self-regulatory self-efficacy ($F = 22.043$, $p < .001$, $\eta^2 = 0.138$), and classroom performance self-efficacy (F = 8.769, $p < .001$, $\eta^2 = 0.06$), and genre-based performance self-efficacy ($F = 10.578$, $p < .001$, $\eta^2 = 0.086$). The statistically significant differences across four profiles were also found in L2 student writers' use of motivation regulation strategies: interest enhancement ($F = 33.566$, $p < .001$, $\eta^2 = 0.196$), motivational self-talks ($F = 64.313$, $p < .001$, $\eta^2 = 0.294$), and emotional control ($F = 39.968$, $p < .001$, $\eta^2 = 0.225$).

Furthermore, to observe the nuanced distinctions between the profile pairs, we performed the post hoc Schéffé's Test on the dimensions of writing self-efficacy and motivation regulation strategies, the results of which are shown in Table 4.

The results of Scheffe's Test in Table 4 reveal that all profile pairs were statistically significantly different in dimensions of writing self-efficacy and motivation regulation strategies rather than the following profile pairs: the LAGs profile versus the MH profile in classroom performance self-efficacy ($p = .756$), the LAGs profile versus the AAGs profile pairs in classroom performance self-efficacy ($p = .337$), the MH profile versus the AAGs profile pair in linguistic self-efficacy ($p = .848$), classroom performance self-efficacy ($p = 1$), and genre-based performance self-efficacy ($p = .416$), the MH profile versus the HAGs profile pair in linguistic self-efficacy ($p = .161$), self-regulatory self-efficacy ($p = .673$), genre-based performance self-efficacy ($p = .375$), interest enhancement ($p = .439$), motivational self-talk ($p = .972$), and emotional control ($p = .999$).

Table 4 Differences in writing self-efficacy and motivation regulation strategies across latent profiles

Factors	Statistics	Profile pair					
		LAGs-MH	LAGs-AAGs	LAGs-HAGs	MH-AAGs	MH-HAGs	AAGs-HAGs
Writing achievement goals							
MG	MD	-2.204	-1.178	-2.258	1.026	−0.054	-1.081
	SE	0.077	0.046	0.072	0.065	0.085	0.059
	$p^{scheffe}$	<0.001	<0.001	<0.001	<0.001	0.939	<0.001
PApG	MD	−0.609	−0.412	-1.790	0.197	-1.181	-1.378
	SE	0.190	0.115	0.179	0.160	0.211	0.147
	$p^{scheffe}$	0.017	0.006	<0.001	0.681	<0.001	<0.001
PAvG	MD	. 861	−0.309	-1.307	-1.171	-2.169	−0.998
	SE	0.199	0.121	0.188	0.168	0.221	0.155
	$p^{scheffe}$	<0.001	0.09	<0.001	<0.001	<0.001	<0.001
Writing self-efficacy							
LS	MD	−0.821	−0.601	-1.551	0.220	−0.732	−0.951
	SE	0.290	0.176	0.273	0.244	0.321	0.224
	$p^{scheffe}$	0.047	0.009	<0.001	0.848	0.161	<0.001
SRS	MD	-1.385	−0.690	-1.727	0.695	−0.342	-1.036
	SE	0.248	0.151	0.234	0.209	0.275	0.192
	$p^{scheffe}$	<0.001	<0.001	<0.001	0.012	0.673	<0.001
CPS	MD	−0.340	−0.349	-1.463	−0.009	-1.123	-1.115
	SE	0.312	0.189	0.294	0.263	0.346	0.242
	Sig.	0.276	0.066	0.000	0.974	0.001	0.000
	$p^{scheffe}$	0.756	0.337	<0.001	1	0.016	<0.001
GPS	MD	-1.013	−0.6201	-1.552	0.392	−0.540	−0.9320
	SE	0.275	0.167	0.259	0.232	0.306	0.213
	$p^{scheffe}$	0.004	0.004	<0.001	0.416	0.375	<0.001
Motivation regulation strategies							
IE	MD	-1.833	-1.026	-2.320	0.807	−0.488	-1.294
	SE	0.267	0.162	0.251	0.225	0.296	0.207
	$p^{scheffe}$	<0.001	<0.001	<0.001	0.005	0.439	<0.001
MST	MD	-2.166	-1.115	-2.281	1.051	−0.115	-1.166
	SE	0.214	0.130	0.202	0.181	0.238	0.166
	$p^{scheffe}$	<0.001	<0.001	<0.001	<0.001	0.972	<0.001
EC	MD	-2.224	-1.214	-2.274	1.011	−0.050	-1.061
	SE	0.259	0.157	0.244	0.218	0.287	0.200
	$p^{scheffe}$	<0.001	<0.001	<0.001	<0.001	0.999	<0.001

Note. MG=mastery goals, PApG=performance-approach goals, PAvG=performance-avoidance goals, LS=linguistic self-efficacy, SRS=self-regulatory self-efficacy, CPS=classroom performance self-efficacy, GPS=genre-based performance self-efficacy, IE=interest enhancement, MST=motivational self-talk, EC=emotional control, MD=mean difference, SD=standard deviation. *P*-value adjusted for a family of 4, LAGs=Low All Goals, MH=Mastery High, AAGs=Average All Goals, HAGs=High All Goals

Profile differences in writing achievements

L2 student writers' differences in writing achievements in the argumentative task and the narrative task across the identified profiles of writing achievement goals were also examined utilizing ANOVA. The results in Table 5 indicate that there are no statistically significant

Table 5 ANOVA results of differences in writing achievements across latent profiles

	Mean				Levene (df)	F(df)	Sig.	η^2
	LAGs	MH	AAGs	HAGs				
Argumentation	74.000	73.125	75.382	76.313	0.224(3,412)	ANOVA F (3,412) = 1.294	0.276	0.009
Narration	70.341	72.800	73.267	74.771	1.210(3,412)	ANOVA F (3,412) = 3.813	0.010	0.027

Note. LAGs = Low All Goals, MH = Mastery High, AAGs = Average All Goals, HAGs = High All Goals

Table 6 Differences in writing achievements in argumentation and narration across latent profiles

Profile (I)	Profile (J)	Mean difference	SE	$p^{scheffe}$
Argumentation				
LAGs	MH	0.875	1.887	0.975
LAGs	AAGs	-1.382	1.145	0.693
LAGs	HAGs	-2.313	1.778	0.639
MH	AAGs	-2.257	1.593	0.571
MH	HAGs	-3.188	2.095	0.51
AAGs	HAGs	−0.93	1.462	0.939
Narration				
LAGs	MH	-2.459	1.59	0.496
LAGs	AAGs	-2.926*	0.965	0.028
LAGs	HAGs	-4.429*	1.498	0.034
MH	AAGs	-0.467	1.342	0.989
MH	HAGs	-1.971	1.765	0.742
AAGs	HAGs	-1.503	1.232	0.685

Note. P-value adjusted for comparing a family of 4, LAGs = Low All Goals, MH = Mastery High, AAGs = Average All Goals, HAGs = High All Goals

differences in writing achievement in argumentative writing across four profiles ($F = 1.294$, $p = .276$, $\eta^2 = 0.009$), but in narrative writing ($F = 3.813$, $p = .01$, $\eta^2 = 0.027$).

In the same vein, to observe the nuanced distinctions between the profile pairs, we performed the post hoc Scheffe's Test on writing achievements in both the argumentative task and the narrative one, the results of which are shown in Table 6.

The statistics in Table 6 showed that the LAGs profile demonstrated poorer performance than the AAGs profile and the HAGs profile in both writing tasks; the MHs profile demonstrated poorer performance than the AAGs profile and the HAGs profile in both writing tasks; the AAGs profile demonstrated poorer performance than the HAGs profile in both writing tasks, the LAGs profile showed poorer performance than the MHs profile in the narrative task but better performance in the argumentative task. However, the aforementioned differences in the writing achievement in the two genres across the profiles were not fully verified by the post-hoc Scheffe's Test. The results of post hoc Scheffe's Test in Table 6 there were significant differences in the following two profile pairs in L2 student writers' writing achievement in the narrative task: the LAGs profile versus the AAGs profile ($p = .028$) and the LAGs profile-the HAGs Profile ($p = .034$), but not the rest profile pairs.

Table 7 Differences in predictiveness of writing achievement goals, writing self-efficacy, and motivation regulation strategies on writing achievement

Factors	overall		LAGs		MH		AAGs		HAGs	
	β	*p*	β	*p*	β	*p*	β	*p*	β	*p*
Argumentation										
MG	−0.013	0.845	−0.062	0.74	0.565	0.104	0	0.994	−0.179	0.477
PApG	0.014	0.836	0.07	0.743	0.11	0.665	0.014	0.835	−0.065	0.883
PAVG	0.002	0.973	−0.247	0.191	−0.416	0.141	0.014	0.822	−0.382	0.28
LS	0.164	0.01*	−0.441	0.022*	0.749	0.061	0.162	0.024	0.644	0.027*
SRS	0.031	0.665	0.247	0.232	−0.813	0.06	0.056	0.471	−0.216	0.425
CPS	−0.044	0.523	0.334	0.123	1.552	0.022*	−0.08	0.284	−0.486	0.217
GPS	0.085	0.223	−0.22	0.353	-1.233	0.029*	0.144	0.056	0.557	0.128
IE	−0.028	0.679	0.149	0.497	−0.034	0.879	−0.076	0.263	0.096	0.853
MST	−0.066	0.423	−0.584	0.043*	0.707	0.051	−0.025	0.743	0.008	0.989
EC	0.017	0.815	0.589	0.025*	-1.154	0.012*	−0.008	0.904	0.327	0.364
Narration										
MG	0.11	0.111	−0.09	0.662	0.277	0.457	0.079	0.21	−0.109	0.57
PApG	−0.05	0.443	−0.252	0.287	−0.241	0.411	−0.062	0.368	0.66	0.715
PAVG	0.102	0.066	−0.271	0.189	−0.568	0.086	0.149	0.016	−0.313	0.222
LS	0.065	0.305	−0.301	0.142	0.318	0.447	0.079	0.274	0.263	0.45
SRS	−0.021	0.773	0.032	0.886	−0.38	0.401	−0.023	0.775	0.122	0.407
CPS	−0.051	0.464	0.154	0.506	0.98	0.159	−0.046	0.544	−0.544	0.703
GPS	0.016	0.815	0.218	0.4	-1.002	0.098	0.045	0.552	0.129	0.243
IE	0.036	0.599	0.193	0.421	0.13	0.612	−0.001	0.985	−0.288	0.756
MST	0.061	0.459	−0.479	0.124	0.608	0.124	0.099	0.198	0.641	0.64
EC	−0.013	0.858	0.581	0.041*	−0.608	0.184	−0.054	0.453	0.024	0.357

Note. MG = mastery goals, PApG = performance-approach goals, PAvG = performance-avoidance goals, LS = linguistic self-efficacy, SRS = self-regulatory self-efficacy, CPS = classroom performance self-efficacy, GPS = genre-based performance self-efficacy, IE = interest enhancement, MST = motivational self-talk, EC = emotional control, LAGs = Low All Goals, MH = Mastery High, AAGs = Average All Goals, HAGs = High All Goals

Profile differences in predictiveness of writing achievement goals, writing self-efficacy, and motivation regulation strategies on writing achievement

Multiple regression analyses were conducted to identify the differences among profiles in predictive effects of writing achievement goals, writing self-efficacy, and motivation regulation strategies on writing achievement, the results of which are displayed in Table 7.

The statistics in Table 7 reveal that in the argumentative writing, linguistic self-efficacy exposed a significant predictive effect on writing achievement ($\beta = 0.164$, $p = .01$) for the overall sample; linguistic self-efficacy, motivational self-talk, and emotional control showed significant predictive impacts on writing achievement in the argumentative writing for the cohort in the LAGs profile ($\beta_{\text{linguistic self–efficacy}} = -0.441$, $p = .022$; $\beta_{\text{motivational self–talk}} = -0.584$, $p = .043$; $\beta_{\text{emotional control}} = 0.589$, $p = .025$); classroom performance self-efficacy, genre-based performance self-efficacy, and emotion control could significantly predict writing achievements in the argumentative writing ($\beta_{\text{classroom performance self–efficacy}} = 1.552$, $p = .022$; $\beta_{\text{genre–based performance self–efficacy}} = -1.233$, $p = .029$ $\beta_{\textit{emotion control}} = \textit{-1.154}$, $p = .012$) for that in the MH profile; linguistic self-efficacy demonstrated significant predictive effect on writing achievements ($\beta = 0.162$, $p = .024$) for that in the AAGs profile;, linguistic self-efficacy

showcased significant predictive effect on writing achievements ($\beta = 0.644$, $p = .027$) for that in the HAGs profile. In contrast, in narrative writing, only emotional control showed a significant predictive effect on writing achievement ($\beta = 0.581$, $p = .041$) for the cohort in the LAGs profile, whereas the variables involved in this study could not significantly predict writing achievement for the overall sample and the cohorts in the MH profile, the AAGs profile, and the HAGs profile.

Discussion

The current study intended to explore the nexus among achievement goals, self-efficacy, and motivation regulation and their effects on L2 writing achievements by adopting a person-based approach. The results of latent profile analyses revealed that four profiles of achievement goals emerged from the L2 student writers: "LAGs", "MH", "AAGs", and "HAGs". Moreover, results of ANOVA and Welch' Test showed that L2 student writers assigned to these four profiles differed in writing achievement goals (i.e., mastery goals, performance-approach goals, performance-avoidance goals), dimensions of writing self-efficacy (linguistic self-efficacy, self-regulatory self-efficacy, classroom performance self-efficacy, genre-based performance self-efficacy), reported use of motivation regulation strategies (interest enhancement, motivational self-talk, and emotional control), and writing achievement in narrative writing. Furthermore, the post hoc Scheffe's Tests discovered the nuances of differences across profiles that might be possibly neglected by ANOVA and Welch's Test: There were not only differences but also similarities between the profile pairs in terms of the aforementioned variables. In addition, the results of the multiple regression analyses revealed that the profiles varied in the predictive effect of writing achievement goals, writing self-efficacy, and motivation regulation strategies on L2 student writers' writing achievement.

Profiles of writing achievement goals

The first research question explored the possible profiles of achievement goals, including mastery goals, performance-approach goals, and performance-avoidance goals. The current study identified four profiles of achievement goals in the sample of L2 student writers involved: "LAGs", "MH", "AAGs", and "HAGs". This finding might be in great contrast to three profiles that Paul et al. (2021) detected in L1 high school learners: High On All, Average On All, and Low On All. Furthermore, the disparity might also be observed in the proportion of the membership in each profile between the current study and Paul et al.'s (2021). In the current study, the LAGs profile accounted for 9.86%, the MH profile for 4.81%, the AAGs profile for 79.57%, and the HAGs profile for 5.77%. On the contrary, in Paul et al.'s study, about 21% of the participants were assigned to the Low on All profile, 47% to the Average on All profile, and 33% to the High on All profile. These differences might suggest that contextual factors in L2 writing (e.g., language proficiency concerns and unfamiliarity with L2 writing conventions) may influence how achievement goals are endorsed (Church et al., 2001; Greene et al., 2004; Jiang et al., 2014). Such differences underscore the need to consider learning contexts as a critical factor shaping achievement goals. Compared with the L2 writing context, the circumstances surrounding L1 writing may more easily initiate

learners to adhere to high mastery goals because the competence to write in L1 is conceived to be crucial to their academic success and future career promotion (e.g., Graham & Perin, 2007; Preiss et al., 2013).

Four profiles of achievement goals identified in L2 student writers echo the profiles summarized in Wormington and Linnenbrink-Garcia's (2017) meta-analytic synthesis of achievement goals. Specifically, three profiles (i.e., MH, HAGs, and AAGs) were consistent with three of the four most prevalent profiles Wormington and Linnenbrink-Garcia (2017) identified in the available studies concerning profiles of achievement goals. The congruence of these profiles across contexts suggests that patterns of goal adoption are relatively stable across different academic domains, including L2 writing. This cross-contextual similarity supports the generalizability of Wormington and Linnenbrink-Garcia (2017) 's framework and highlights the value of using a person-centered approach in motivation research. Moreover, the L2 context introduces domain-specific challenges such as language anxiety, cultural expectations, and self-efficacy related to linguistic competence, which interact with these goal profiles. However, the recurrence of similar goal combinations points to underlying psychological consistencies in how students organize their motivations around learning, achievement, and evaluation.

To conclude, the parallels between the L2 student writer profiles and those in Wormington and Linnenbrink-Garcia (2017) reinforce the robustness of achievement goal theory across domains. They also emphasize the utility of a person-centered approach in understanding motivation heterogeneity in learners. By recognizing these profiles, educators can better tailor interventions, encouraging mastery goals, minimizing avoidance tendencies, and supporting adaptive performance goals, to enhance L2 student writers' motivation and writing success.

Differences in self-efficacy and motivation regulation across the profiles of achievement goals

The second research question is concerned with whether there are differences in self-efficacy and motivation regulation across the profiles of achievement goals in the L2 student writers. The outcomes of ANOVA and Welch's Test indicated the significant differences across profiles of achievement goals with respect to dimensions of writing self-efficacy and motivation regulation strategies. L2 student writers involved in the HAGs profile and the MH profile held higher writing self-efficacy than those in the AAGs profile and the LAGs profile. This result might be interpreted as a function of the positive effect of mastery goals and performance-approach goals on writing self-efficacy revealed by the previous studies (e.g., Limpo & Aves, 2017; Yilmaz Soylu et al., 2017). High mastery goals and performance-approach goals may require learners to channel their engagement, effort, and time into strengthening their writing competence and upgrading their writing achievement. Naturally, student writers endorsing the HAGs profile might more actively and willingly sustain their persistence and passion throughout the plateau, take on new challenges, and overcome obstacles and setbacks, enabling them to establish strong writing self-efficacy to complete specific writing tasks.

Besides, these findings might offer compelling evidence supporting the adaptive effect of the HAGs profile and the MH profile. Student writers in these groups reported greater writing self-efficacy and more frequent use of motivation regulation strategies, which are key

predictors of L2 writing. These profiles are characterized by strong mastery goals, which have been consistently linked to deeper engagement and persistence in challenging L2 writing tasks (Zhang & Zhang, 2025). In contrast, the results highlight the maladaptive effect of the AAGs profile and the LAGs profile. Student writers in these profiles exhibited relatively lower writing self-efficacy and limited use of motivational regulation strategies.

Crucially, these findings challenge the validity of one-size-fits-all approach to writing instruction. Traditional writing pedagogies often assume a uniform set of motivation characteristics across learners and apply generalized strategies accordingly. However, the distinct achievement goals profiles uncovered in this study underscore the heterogeneity of L2 student writers' motivational configurations. What motivates and supports a student in the HAGs profile may not work for another student in the AAGs or LAGs profile. Therefore, instruction that is tailored to students' motivational profiles, for example, by strengthening mastery goals in the LAGs and AAGs profiles, and reinforcing self-efficacy across profiles, may be far more effective than generalized interventions.

Differences in L2 writing achievement across the profiles of achievement goals

The third research question examined whether there are differences in L2 writing achievement across the profiles of achievement goals. The results of ANOVA also revealed that the profiles of achievement goals are significantly different in the writing achievement in the narrative task rather than in that in the argumentative task. This discovery might be interpreted as a function of genre in L2 writing. Research revealed that L2 student writers showed significantly different achievements in both argumentative and narrative tasks (e.g., Hyland, 2004; Yoon & Polio, 2017).

The pair-wise comparison showed that the HAGs profile performed better than the AAGs profile, followed by the MH profile and the LAGs in both writing tasks. These findings concur with those of Troia et al. (2012), who identified three motivational profiles of student writers: highly motivated, average-motivated, and low-motivated, and revealed that both low- and average-motivated writer profiles exhibited slightly lower performance in writing than the highly-motivated writer profile. The convergence between Troia et al. (2012)'s study and this study lies in the hierarchical pattern of writing performance across motivation profiles. Specifically, both studies show that student writers with strong, adaptive motivation profiles (e.g., the HAGs and MH profiles in this study vs. highly motivated profiles in Troia et al. (2012)) perform best in writing; students with mixed or ambivalent motivational profiles (e.g., AAGs profile in this study vs. average-motivated profile in Troia et al. (2012)) tend to show moderate performance outcomes; students with low motivation profiles are least successful in writing tasks, likely due to a lack of persistence, engagement, and confidence. Thus, the findings of the current study corroborate those of Troia et al. (2012) in demonstrating that higher-quality writing performance is closely tied to stronger, more adaptive motivation profiles.

Our findings also echoed the result revealed in the meta-analysis conducted by Wormington and Linnenbrink-Garcia (2017), which states that the all-high profile is correlated with a set of adaptive outcomes. Besides, as argued by Wormington and Linnenbrink-Garcia (2017), examining achievement goal profiles in relation to achievements may allow us to discern the adaptive and maladaptive mixture of achievement goal endorsement. Therefore,

these results might implicate that the HAGs profile is the most adaptive, whereas the LAGs profile is the most maladaptive, with the other two profiles lying in between.

Differences in predictive effects on writing achievements across the profiles of achievement goals

The fourth research question examined whether the predictive effects of writing achievement goals, writing self-efficacy, and motivation regulation strategies on L2 writing achievement would vary across different profiles. The outcomes of multiple regression analyses showed that linguistic self-efficacy could significantly predict the writing achievement in the argumentative writing of the whole sample of L2 student writers involved in the current study, thus leading further evidence to corroborate the importance of linguistic self-efficacy for L2 student writers (e.g., Teng et al., 2018; Zhang & Zhang, 2023).

However, multiple regression analyses conducted on profiles of achievement goals revealed the unnoticeable nuance of the predictive effect of writing self-efficacy and motivation regulation strategies. Specifically, for learners belonging to the HAGs profile, their linguistic self-efficacy, motivational self-talk, and emotional control could significantly predict their writing achievement in the argumentative writing; for those involved in the MH profile, their classroom performance self-efficacy, genre-based performance self-efficacy, and emotional control exerted significant impacts on their writing achievement in the argumentative writing. In contrast, for those engaging in both the AAGs profile and the LAGs profile, linguistic self-efficacy served as a significant predictor of their writing achievement in argumentative writing. These findings might suggest the relative significance of writing self-efficacy and motivation regulation strategies for L2 student writers involved in different profiles of achievement goals, contributing to the nuanced understanding of the influences of these aforementioned variables on L2 writing achievement, which could be overlooked in the previous research endeavors based on the variable-based approach.

Interesting, in narrative writing, only emotional control was significantly predictive of writing achievement for the LAGs profile, while the aforementioned predictive effects in argumentative writing did not show up in narrative writing. These results suggested that task type (i.e., genre) may moderate how motivational constructs impact L2 writing achievement. The differences might be attributed to L2 student writers' familiarity with argumentative writing rather than narrative writing. Compared with narrative writing, argumentative writing has been practiced more frequently by L2 student writers involved in this study. These nuances underscore that a one-size-fits-all model of motivational support is insufficient; rather, targeted interventions may be needed based on the specific motivational profile of L2 student writers.

Conclusion

The current study explored the links among achievement goals, writing self-efficacy, and motivation regulation strategies in the population of L2 student writers and their influences on L2 student writers' writing achievement by adopting the person-based approach. The results of latent profile analyses revealed that four profiles of achievement goals emerged from the L2 student writers involved in the current study: "LAGs", "MH", "AAGs", and

"HAGs". Moreover, the results of ANOVA and Welch's Test showed that L2 student writers assigned to these four profiles differed in writing achievement goals, dimensions of writing self-efficacy, reported use of motivation regulation strategies, and writing achievement in the narrative task rather than that in the argumentative task. In addition, the results of the multiple regression analyses revealed that the profiles varied in the predictive effect of writing achievement goal, writing self-efficacy, and motivation regulation strategies on L2 writing achievement.

These findings might offer some implications for research and pedagogical practice in L2 writing and, more broadly, L2 learning. Methodologically, the use of a person-centered approach enables researchers to detect group-level heterogeneity in L2 student writers with regard to key individual differences—such as language aptitude, working memory, motivation, metacognition, self-beliefs, emotion, and learning strategies—by identifying distinct profiles of achievement goals. This approach not only reveals how these variables cluster within individuals but also clarifies the coherence among these constructs within each profile, offering a more nuanced understanding of L2 student writers' motivational configuration.

Pedagogically, the identification of distinct profiles of achievement goals signals that L2 learners do not operate with a single, uniform set of motivational beliefs. Instead, they typically embody a complex blend of mastery, performance-approach, and performance-avoidance goals, which relate to self-efficacy and motivation regulation strategies in multifaceted ways. This complexity cautions against interventions that focus solely on one aspect of motivation. Instead, L2 writing instructors should adopt a multifaceted approach that acknowledges and addresses the interplay among achievement goals, self-efficaccy, and motivation regulation strategies. For example, teaching strategies should be tailored so that (1) student writers in lower achievement goal profiles (AAGs and LAGs) receive targeted support to boost their overall motivational orientation—such as enhancing linguistic self-efficacy and encouraging the adoption of motivation regulation strategies—to elevate their engagement and writing performance; (2) student writers in the MH profile benefit from instructional practices that help them further develop their performance-approach goals, thereby leveraging the potential of balancing high mastery goals with medium performance-approach goals; (3) student writers in the HAGs profile are provided with opportunities to sustain and refine their high achievement goals, strong motivation regulation, and writing self-efficacy through advanced challenges and personalized feedback. Moreover, the differential predictive effects observed across profiles suggest that educators should design profile-specific instructional interventions. For instance, while enhancing linguistic self-efficacy might be especially critical for student writers in the AAGs and LAGs profiles, fostering refined emotional control and targeted self-regulatory strategies might yield greater benefits for those in HAG profiles.

Admittedly, several limitations should be noted. Firstly, the participants involved in the current study were mainly beginning and intermediate L2 student writers rather than advanced ones. Accordingly, the findings of the current study cannot be generalized to advanced L2 student writers. Therefore, it might be recommended that future studies entail advanced L2 student writers to verify or falsify the findings of the current study. Secondly, only two writing tasks were employed to measure the participants' writing achievements. Therefore, to grasp the full landscape of the differences across the profiles of achievement goals concerning predictive effects of writing achievement goals, writing self-efficacy,

and motivation regulation strategies on L2 writing achievement, it is advised that more writing tasks (i.e., descriptive and expositive tasks) should be included. Thirdly, this study employed several self-report scales to measure the participants' self-perceived writing achievement goals, writing self-efficacy, and motivation regulation strategies, respectively. Because of the participants' memory bias and social desirability bias, the self-report scales might elicit unreliable or inaccurate responses from the participants. Accordingly, it might be advisable to adopt the data from other resources (e.g., observation, interview, and think-aloud) to cross-validate self-reported data in future research. Finally, the writing tasks were arranged in a fixed order. It is recommended that the researcher employ counterbalancing techniques to mitigate the possible order effect in future studies. Finally, although the analytical methods employed, especially latent profile analysis and ANOVA, were appropriate for answering the research questions, they are primarily exploratory and descriptive. While suitable for uncovering patterns and group differences, they do not test the mechanisms or causal pathways through which achievement goals influence L2 writing performance. Incorporating structural equation modeling or path analysis in future work could help clarify the mediating or moderating roles of writing self-efficacy, motivation regulation strategy use, and other motivation processes.

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Declarations

Human ethics The study was approved by The University of Auckland Ethics Committee on Researching Human Participants (Protocol Number: UAHEC21641).

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References

Bandura, A. (1999). Social cognitive theory: An agentic perspective. *Asian Journal of Social Psychology*, *2*(1), 21–41. https://doi.org/10.1111/1467-839x.00024

Bandura, A. (2006). Toward a psychology of human agency. *Perspectives on Psychological Science*, *1*(2), 164–180. https://doi.org/10.1111/j.1745-6916.2006.00011.x

Bandura, A. (2018). Toward a psychology of human agency: Pathways and reflections. *Perspectives on Psychological Science*, *13*(2), 130–136. https://doi.org/10.1177/1745691617699280

Barron, K. E., & Harackiewicz, J. M. (2001). Achievement goals and optimal motivation: Testing multiple goal models. *Journal of Personality and Social Psychology*, *80*(5), 706–722. https://doi.org/1.1037/0022-3514.8.5.706.

Boekaerts, M., & Cascallar, E. (2006). How Far have we moved toward the integration of theory and practice in self-regulation? *Educational Psychology Review*, *19*, 199–210.

Bruning, R., Dempsey, M. S., Kauffman, D. F., McKim, C., & Zumbrunn, S. (2013). Examining dimensions of self-efficacy for writing. *Journal of Educational Psychology*, *105*, 25–38.

Camacho, A., De Smedt, F., Maio, R., Cadima, J., Van Keer, H., & Alves, R. A. (2023). Writing motivation profiles and their association with writing performance: A person-centered approach. *Topics in Language Disorders*, *43*(2), 169–186. https://doi.org/10.1097/TLD.0000000000000314

Chea, S. (2012). *The relationships among writing self-efficacy, writing goal orientation, learning strategies, and writing achievement* (Master Thesis). Available from ProQuest Dissertations and Theses database. (UMI 1513414).

Church, M. A., Elliot, A. J., & Gable, S. L. (2001). Perceptions of classroom environment, achievement goals, and achievement outcomes. *Journal of Educational Psychology, 93*, 43–54. https://doi.org/1.1037/0022-0663.93.1.43.

Conley, A. M. (2012). Patterns of motivation beliefs: Combining achievement goal and expectancy-value perspectives. *Journal of Educational Psychology*, *104*(1), 32–47. https://doi.org/1.1037/a0026042.

Dörnyei, Z., & Taguchi, T. (2010). *Questionnaires in second language research: Construction, administration, and processing* (2nd ed.). Routledge.

Dweck, C. S., & Leggett, E. L. (1988). A social-cognitive approach to motivation and personality. *Psychological Review*, *95*(2), 256–273. https://doi.org/1.1037/0033-295X.95.2.256.

Elliot, A. J. (1999). Approach and avoidance motivation and achievement goals. *Educational Psychologist*, *34*(3), 169–189. https://doi.org/1.1207/s15326985ep3403_3.

Elliot, A. J., & Church, M. A. (1997). A hierarchical model of approach and avoidance achievement motivation. *Journal of Personality and Social Psychology*, *72*(1), 218–232. https://doi.org/1.1037/0022-3514.72.1.218

Elliot, A. J., & Murayama, K. (2008). On the measurement of achievement goals: Critique, illustration, and application. *Journal of Educational Psychology*, *100*(3), 613–628. https://doi.org/1.1037/0022-0663.10.3.613.

Graham, S., Harris, K. R., & Mason, L. (2005). Improving the writing performance, knowledge, and self-efficacy of struggling young writers: The effects of self-regulated strategy development. *Contemporary Educational Psychology*, *30*(2), 207–241.

Graham, S., & Perin, D. (2007). A meta-analysis of writing instruction for adolescent students. *Journal of Educational Psychology*, 99(3), 445–476. https://doi.org/1.1037/0022-0663.99.3.445.

Greene, B. A., Miller, R. B., Crowson, H. M., Duke, B. L., & Akey, K. L. (2004). Predicting high school students' cognitive engagement and achievement: Contributions of classroom perceptions and motivation. *Contemporary Educational Psychology*, *29*, 462–482. https://doi.org/1.1016/j.cedpsych.2004.01.006

Harackiewicz, J. M., Barron, K. E., & Elliot, A. J. (1998). Rethinking achievement goals: When are they adaptive for college students and why? *Educational Psychologist*, *33*(1), 1–21. https://doi.org/1.1207/s15326985ep3301_1.

Harackiewicz, J. M., Barron, K. E., Pintrich, P. R., Elliot, A. J., & Thrash, T. M. (2002). Revision of achievement goal theory: Necessary and illuminating. *Journal of Educational Psychology*, *94*(3), 638–645. https://doi.org/1.1037//0022-0663.94.3.638

Hashemian, M., & Heidari, A. (2013). The relationship between L2 learners' motivation/attitude and success in L2 writing. *Procedia - Social and Behavioral Sciences*, *70*, 476–489. https://doi.org/10.1016/j.sbspro.2013.01.085

Huang, C. (2011). Achievement goals and achievement emotions: A meta-analysis. *Educational Psychology Review*, *23*, 359–388. https://doi.org/1.1007/s10648-011-9155-x

Huang, C. (2016). Achievement goals and self-efficacy: A meta-analysis. *Educational Research Review*, *19*, 119–137. https://doi.org/1.1016/j.edurev.2016.07.002

Hulleman, C. S., Schrager, S. M., Bodman, S. M., & Harackiewicz, J. M. (2010). A meta-analytic review of achievement goal measures: Different labels for the same constructs or Different constructs with similar labels? *Psychological Bulletin*, *136*, 422–449.

Hyland, K. (2004). *Genre and second language writing*. University of Michigan Press.

Jacobs, H. L., Zingraf, S. A., Wormuth, D. R., Hartfiel, V. F., & Hughey, J. B. (1981). *Testing ESL composition: A practical approach*. *English Composition Program*. Newbury House. https://eric.ed.gov/?id=ed217708

Jiang, Y., Song, J., Lee, M., & Bong, M. (2014). Self-efficacy and achievement goals as motivational links between perceived contexts and achievement. *Educational Psychology*, *34*(1), 92–117. https://doi.org/1.1080/0144341.2013.863831.

Jung, T., & Wickrama, K. A. S. (2008). An introduction to latent class growth analysis and growth mixture modeling. *Social and Personality Psychology Compass, 2*(1), 302–317. https://doi.org/1.1111/j.1751-9004.2007.00054.x.

Kaplan, A., Lichtinger, E., & Gorodetsky, M. (2009). Achievement goal orientations and self-regulation in writing: An integrative perspective. *Journal of Educational Psychology*, *101*(1), 51–69. https://doi.org/1.1037/a0013200.

Lee, M., & Evans, M. (2019). Investigating the operating mechanisms of the sources of L2 writing self-efficacy at the stages of giving and receiving peer feedback. *The Modern Language Journal*, *103*(4), 831–847. https://doi.org/10.1111/modl.12598

Limpo, T., & Alves, R. A. (2017). Relating beliefs in writing skill malleability to writing performance: The mediating role of achievement goals and self-efficacy. *Journal of Writing Research*, *9*(2), 97–125. https://doi.org/1.17239/jowr-2017.09.02.01.

Linnenbrink-Garcia, L., Tyson, D. F., & Patall, E. A. (2008). When are achievement goal orientations beneficial for academic achievement? A closer look at moderating factors. *Revue Internationale De Psychologie Sociale*, *21*(1–2), 19–70. https://psycnet.apa.org/record/2014-30636-002

MacArthur, C. A., Philippakos, Z. A., & Graham, S. (2016). A multicomponent measure of writing motivation with basic college writers. *Learning Disability Quarterly*, *39*(1), 31–43. https://doi.org/10.1177/0731948715583115

Maehr, M. L., & Zusho, A. (2009). Achievement goal theory: The past, present, and future. In K. Wentzel (Ed.), *Handbook of motivation at school* (pp. 77–104). Routledge.

Manchón, R. M., Roca de Larios, J., & Murphy, L. (2007). A review of writing strategies: Focus on conceptualization and impact of the first Language. In A. D. Cohen, & E. M. Macaro (Eds.), *Language learner strategies: Thirty years of research and practice* (pp. 229–250). Oxford University Press.

Middleton, M. J., & Midgley, C. (1997). Avoiding the demonstration of lack of ability: An underexplored aspect of goal theory. *Journal of Educational Psychology*, *89*, 710–718. https://doi.org/1.1037/0022-0663.89.4.710

Muthen, L. K., & Muthen, B. (2017). *Mplus user's guide: Statistical analysis with latent variables* (8th ed.). Muthén & Muthén.

Niemivirta, M., Pulkka, A. T., Tapola, A., & Tuominen, H. (2019). Achievement goal orientations. In K. A. Renninger & S. E. Hidi (Eds.), *The Cambridge handbook of motivation and learning* (pp. 566–616). Cambridge University Press. https://doi.org/1.1017/9781316823279.025.

Pajares, F., Britner, S., & Valiante, G. (2000). Relation between achievement goals and self-beliefs of middle school students in writing and science. *Contemporary Educational Psychology*, *25*(4), 406–422. https://doi.org/1.1006/ceps.1999.1027.

Pajares, F., & Cheong, Y. F. (2003). Achievement goal orientations in writing: A developmental perspective. *International Journal of Educational Research*, *39*(4–5), 437–455. https://doi.org/1.1016/j.ijer.2004.06.008.

Paul, N., Lin, T. J., Ha, S. Y., Chen, J., & Newell, G. (2021). The role of achievement goal orientations in the relationships between high school students' anxiety, self-efficacy, and perceived use of revision strategies in argumentative writing. *Journal of Writing Research*, *12*(3), 657–684. https://doi.org/1.17239/jowr-2021.12.03.05.

Pintrich, P. (2004). A conceptual framework for assessing motivation and self-regulated learning in college students. *Educational Psychology Review*, *16*, 385–407.

Pintrich, P. R. (2000). The role of goal orientation in self-regulated learning. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 451–502). Academic.

Preiss, D. D., Castillo, J. C., Grigorenko, E. L., & Manzi, J. (2013). Argumentative writing and academic achievement: A longitudinal study. *Learning and Individual Differences*, *28*, 204–211. https://doi.org/1.1016/j.lindif.2012.12.013.

Schwinger, M., Steinmayr, R., & Spinath, B. (2009). How do motivational regulation strategies affect achievement: Mediated by effort management and moderated by intelligence. *Learning and Individual Differences*, *19*, 621–627. http://dx.doi.org/1.1016/j.lindif.2009.08.006

Schwinger, M., & Wild, E. (2012). Prevalence, stability, and functionality of achievement goal profiles in mathematics from third to seventh grade. *Contemporary Educational Psychology*, *37*(1), 1–13. https://doi.org/1.1016/j.cedpsych.2011.08.001.

Senko, C., Hulleman, C. S., & Harackiewicz, J. M. (2011). Achievement goal theory at the crossroads: Old controversies, current challenges, and new directions. *Educational Psychologist*, *46*(1), 26–47. https://doi.org/1.1080/0046152.2011.538646.

Teng, L. S. (2021). Individual differences in self-regulated learning: Exploring the nexus of motivational beliefs, self-efficacy, and SRL strategies in EL2 writing. *Language Teaching Research*, 38. https://doi.org/1.1177/13621688211006881

Teng, L. S., & Zhang, J. (2018). Effects of motivational regulation strategies on writing performance: A mediation model of self-regulated learning of writing in English as a second/foreign language. *Metacognition and Learning*, *13*(2), 213–24. https://doi.org/1.1007/s11409-017-9171-4.

Teng, L. S., & Zhang, J. (2020). Empowering learners in the second/foreign Language classroom: Can self-regulated learning strategies-based writing instruction make a difference? *Journal of Second Language Writing*, *48*, 100701. https://doi.org/10.1016/j.jslw.2019.100701

Teng, L. S., & Zhang, L. J. (2022). Can self-regulation be transferred to second/foreign language learning and teaching? Current status, controversies, and future directions. *Applied Linguistics*, *43*(3), 587–595. https://doi.org/10.1093/applin/amab032

Troia, G. A., Shankland, R. K., & Wolbers, K. A. (2012). Motivation research in writing: Theoretical and empirical considerations. *Reading & Writing Quarterly*, *28*(1), 5–28. https://doi.org/10.1080/10573569.2012.632729

Tuominen-Soini, H., Salmela-Aro, K., & Niemivirta, M. (2008). Achievement goal orientations and subjective well-being: A person-centred analysis. *Learning and Instruction*, *18*(3), 251–266. https://doi.org/1.1016/j.learninstruc.2007.05.003.

Tuominen-Soini, H., Salmela-Aro, K., & Niemivirta, M. (2011). Stability and change in achievement goal orientations: A person-centered approach. *Contemporary Educational Psychology*, *36*(2), 82–10. https://doi.org/1.1016/j.cedpsych.201.08.002.

Wigfield, A., & Eccles, J. S. (Eds.). (2002). *Development of achievement motivation*. Academic.

Wolters, C. (2003). Regulation of motivation: Evaluating an underemphasized aspect of self-regulated learning. *Educational Psychologist*, *38*, 189–205.

Wolters, C. (2004). Advancing achievement goal theory: Using goal structures and goal orientations to predict students' motivation, cognition, and achievement. *Journal of Educational Psychology*, *96*, 236–225. http://dx.doi.org/1.1037/0022-0663.96.2.236

Wang, K., Zhang, L. J., & Cooper, M. (2025). Taking stock of metacognitive strategies and collaborative writing for EFL learners' writing development. *Journal of Second Language Writing, 68*， 101211. https://doi.org/10.1016/j.jslw.2025.101211

Wolters, C., & Rosenthal, H. (2000). The relation between students' motivational beliefs and their use of motivational regulation strategies. *International Journal of Educational Research, 33*(7–8), 801–82. https://doi.org/1.1016/S0883-0355(00)00051-3.

Wormington, S. V., & Linnenbrink-Garcia, L. (2017). A new look at multiple goal pursuit: The promise of a person-centered approach. *Educational Psychology Review*, *29*(3), 407–445. https://doi.org/1.1016/S0883-0355(00)00051-3https://doi.org/1.1007/s10648-016-9358-2.

Yang, C., & Zhang, L. J. (2023). *Think-aloud protocols in second language writing: A mixed-methods study of their reactivity and veridicality*. Springer Nature. https://doi.org/10.1007/978-3-031-39574-1

Yilmaz Soylu, M., Zeleny, M. G., Zhao, R., Bruning, R. H., Dempsey, M. S., & Kauffman, D. F. (2017). Secondary students' writing achievement goals: Assessing the mediating effects of mastery and performance goals on writing self-efficacy, affect, and writing achievement. *Frontiers in Psychology*, *8*, 1–11. https://doi.org/1.3389/fpsyg.2017.01406.

Yoon, H.J., & Polio, C. (2017). The linguistic development of students of English as a second language in two written genres. *TESOL Quarterly*, *51*(2), 275–301. https://doi.org/10.1002/tesq.296

Zhang, D., & Zhang, L.J. (2019). Metacognition and self-regulated learning in second/foreign language teaching. In X. A. Gao (Ed.), *Second handbook of English language teaching* (pp. 883–898). Springer Nature. https://doi.org/10.1007/978-3-030-02899-2_47

Zhang, J. (2023). Modelling the interplay of writing achievement goals and grit in predicting L2 writing achievements. *System*, *117*, 103118. https://doi.org/10.1016/j.system.2023.103118

Zhang, J., & Zhang, L.J. (2023). Examining the relationship between English as a foreign language learners' cognitive abilities and L2 grit in predicting their writing performance. *Learning and Instruction*, *88*, 101808. https://doi.org/10.1016/j.learninstruc.2023.101808

Zhang, J., & Zhang, L. J. (2025). Exploring the profiles of foreign language learners' writing self-regulation: focusing on individual differences. *Reading and Writing*, *38*, 1533–1569. https://doi.org/10.1007/s11145-024-10568-x

Zhang, J., Zhang, L. J., & Zhu, Y. (2023). Development of a genre-based second language (L2) writing self-efficacy scale. *Frontiers in Psychology*, *14*, 1181196, 1–13. https://doi.org/10.3389/fpsyg.2023.1181196

Zhang, L.J. (2022). L2 writing: Toward a theory-practice praxis. In E. Hinkel (Ed.), *Handbook of practical second language teaching and learning* (pp. 331–343). New York, NY: Routledge. https://doi.org/10.4324/9781003106609-27

Zhang, L. J., Fathi, J., Mohammad Hosseini, H., Derakhshesh, A., & Mehraein, S. (2025). Investigating the role of ideal L2 writing self, writing growth mindset, and writing enjoyment in L2 writing self-efficacy: A mediation model. *Applied Linguistics Review*. https://doi.org/10.1515/applirev-2023-0247

Zimmerman, B. J. (1989). A social cognitive view of self-regulated academic learning. *Journal of Educational Psychology, 81*(3), 329–339. https://doi.org/10.1037/0022-0663.81.3.329

Zimmerman, B. J. (2000). Attaining self-regulation: A social cognitive perspective. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation: Theory, research and applications* (pp. 13–39). Academic.

Zimmerman, B. J., & Risemberg, R. (1997). Becoming a self-regulated writer: A social cognitive perspective. *Contemporary Educational Psychology*, *22*, 73–101. https://doi.org/1.1006/ceps.1997.0919

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